2.3.1.1 Dyeing at Fibre Stage

The process of dyeing at fibre stage is called as fibre dyeing. It includes stock dyeing, top dyeing and dope dyeing.

Stock Dyeing

Fibres are dyed by the process of stock dyeing. In this method, dyeing is carried out in a large enclosed vessel called kiers. Fibres are placed loosely in perforated containers and kept inside the kiers. The dye liquor is circulated through the fibres at high temperature until the desired colour is obtained. Excess dye solution is removed after dyeing process. The dyed fibres are washed and dried.



Figure 2.1 Stock Dyeing

Advantages

- Produces varied colour effects by blending different coloured fibres
- Large quantities of fibres can be dyed at one time
- Produces uniform colour
- Colour fastness ranges from good to excellent
- Dye easily penetrates the fibres and hence crocking is prevented.

Disadvantages

- Expensive and time consuming
- Fibre wastage if not consumed immediately
- Dyed fibres loses its flexibility and hence not readily spin as undyed fibre
- Cannot withstand rapid changes in fashion

Top Dyeing

Wool that has been combed to remove short fibres is called as **top**. Top is wound on perforated spools and dye solution is passed through it. Top dyeing method results in even dyeing.



Figure 2.2 Top Dyeing

Advantages

- Uniform dye uptake
- Dye penetrates into the fibre therefore good colour fastness

Disadvantages

- Flexibility is reduced
- Time consuming

Dope dyeing

Manmade fibres such as polyester and polypropylene are dyed by this method. In this method, dye is mixed with the spinning solution before the filament is extruded through the spinneret.

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